Introduction to Demographic Methods

(3: Fertility)

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1 Introduction. Why study Demography? . . . Git, version control, course materials GitHub
2 Mortality
3 Fertility
4 Population projections
5 Migration
~~6 Kinship~~
7 The future, and Digital and Computational Demography
8 Wrap up + Description of examination
9 Final examination: 3 min presentation to share your selected topic + 2 min Q&A; 6 weeks to prepare a short essay and empirical analysis

Discussing previous week's materials

1) Slides, Readings

1. Any questions on my slides?
2. Let's discuss the "key reading material".
3. Example: Discuss at least 1-2 points, e.g., 1) What was surprising and counterintuitive for you? 2) What did you find that was already expected? 3) What could be extended in this reading/work?
4. Any thoughts about the additional reading materials?

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2) Hands-on

1. Any problems in installing the requirements?
2. Any issues in running the codes in hands-on materials to reproduce the examples?
3. Did anyone extend the examples? How? What did you do and find?

Few points

- Clarification on ax in mortality topic and life table. It is the average years contributed to person years in each time period. For 5-year periods, we use 2.5 and so on.
- MAlexander slides, "4_fertility",
- P4, potential reasons for fertility changes and demographic transitions
- P38: In any given year, fertility rates can decline because of two reasons (Wachter, 2014):
  - Quantum: decline in number of children
  - Tempo: shift in age at childbearing (delaying having kids to older ages)
- Tempo-adjusted TFR allows considering mean age at childbearing in period t to address this effect on period fertility as the cohort measure is not affected.

Switch to Ernesto Amaral's slides: "Lecture04" and Monica Alexander's slides: "4_fertility"

Hands-on part

Needed installations (or checking requirements)

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For today's hands-on part, see the Readme file under "03_MonicaALabs" folder and download the requested files under "data" folder.

We will go over the notebook called "4_fertility_edits.qmd" and see what it does. You need to activate Renv and install R packages listed in the first code block of the notebook, i.e., "tidyverse, here, readxl, janitor" (please check the notebook for the latest list).

Needed: R, RStudio, and Renv environment.

Where to learn basics of R and Python?

To start with Python

Check this website first:
https://www.w3schools.com/python/python_intro.asp

Check this repository by Vincent Traag and others, for an introductory course and code:
https://github.com/vtraag/intro-python

Or this one by Data Carpentry:
https://datacarpentry.org/python-ecology-lesson/

To start with R

Check this website first:
https://www.w3schools.com/r/default.asp

This "very short introduction to R" is a good start:
https://cran.r-project.org/doc/contrib/Torfs+Brauer-Short-R-Intro.pdf

Or this course by Data Carpentry:
https://datacarpentry.org/R-genomics/index.html

Reading materials for today

Key reading material, Chapters 4 and 6

Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.

Additional reading materials

Alexander, M. (2026). Advanced Data Analysis course in Demographic Methods. Retrieved April 8, 2026, from https://github.com/MJAlexander/soc6708

Preston, S., Heuveline, P., & Guillot, M. (2000). Demography: Measuring and Modeling Population Processes. Wiley.

What to do before the next session?

1) Slides, Readings

1. Study my slides carefully.
2. Check the slide with reading materials.
3. Read the “key reading material”.
4. Prepare at least 1-2 discussion points, e.g., What was surprising and counterintuitive for you? What did you find that was already expected? What could be extended in this reading/work?
5. Skim over the additional materials.

2) Hands-on

1. Check the slide and folder for hands-on materials.
2. Install the requirements as outlined.
3. Run the codes in hands-on materials to reproduce the examples, and if you can extend them.
4. Note down the unclear points, errors, and questions for the next session.

Thanks for your attention!

Questions and comments are always welcome!

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LinkedIn/BlueSky/Twitter/Mastodon: @akbaritabar

References

Wachter, K. W. (2014, June). Essential Demographic Methods. Harvard University Press.